

Theory of Machines

BRANCH: MECHANICAL ENGG. (ME)

COURSE CODE: ME301

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Theory of Machines (ME301) under the Mechanical Engg. (ME) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Mechanisms & Kinematics

Link, kinematic pair, kinematic chain, mechanism, degrees of freedom, Grubler's and Kutzbach criteria. Inversions of four-bar chain, slider-crank, and double slider-crank mechanisms.

Unit 2: Velocity & Acceleration

Relative velocity method, instantaneous center method, Coriolis acceleration component, Klein's construction.

Unit 3: Cams, Gears & Gear Trains

Cam profiles for uniform velocity, SHM, and cycloidal motion. Gears: classification, law of gearing, interference, minimum number of teeth. Simple, compound, reverted, epicyclic gear trains.

Unit 4: Governors, Gyroscopes & Balancing

Centrifugal governors: Watt, Porter, Proell, Hartnell. Gyroscopic couple, effect on aircrafts and ships. Static and dynamic balancing of rotating masses.

*Grubler's Criteria: $F = 3(L - 1) - 2J - H$. Gyroscopic Couple: $C = I * \omega * \omega_p$.*

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers



